Epidemic dynamics and paired arrival-time shift by community size

Full Micro
MicroMacro (independent RNG)
$n = 50$, random-connected, $d = 10.0$, $L = 10$, $k = 1$, $\beta = 0.003$, $T_{arr,max} = 20000$, $T_{traj} \leq 4000$, $N = 100$
$n = 100$, random-connected, $d \approx 10.0$, $L = 10$, $k = 1$, $\beta = 0.003$, $T_{arr,max} = 20000$, $T_{traj} \leq 4000$, $N = 100$
$n = 250$, random-connected, $d \approx 10.0$, $L = 10$, $k = 1$, $\beta = 0.003$, $T_{arr,max} = 20000$, $T_{traj} \leq 4000$, $N = 100$
$n = 1000$, random-connected, $d \approx 10.0$, $L = 10$, $k = 1$, $\beta = 0.003$, $T_{arr,max} = 20000$, $T_{traj} \leq 20000$, $N = 100$
$n = 50$, random-connected, $d \approx 10.0$, $L = 10$, $k = 5$, $\beta = 0.003$, $T_{arr,max} = 20000$, $T_{traj} \leq 4000$, $N = 100$
$n = 100$, random-connected, $d \approx 10.0$, $L = 10$, $k = 5$, $\beta = 0.003$, $T_{arr,max} = 20000$, $T_{traj} \leq 4000$, $N = 100$
$n = 250$, random-connected, $d \approx 10.0$, $L = 10$, $k = 5$, $\beta = 0.003$, $T_{arr,max} = 20000$, $T_{traj} \leq 4000$, $N = 100$
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total infected $I(t)$
time
$T_i^{MM} - T_i^{Micro}$
community index